KAIZEN MATHS

Lesson starter

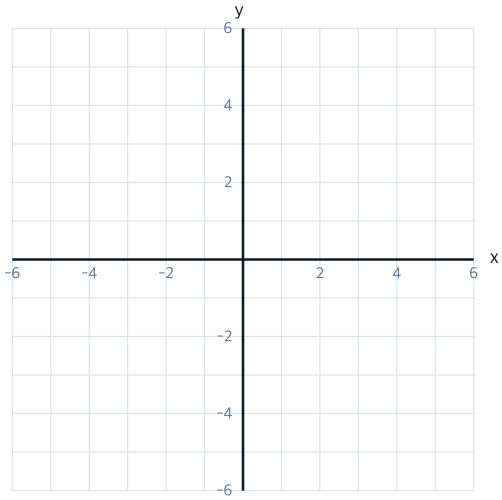
Graph Transformations and Curve Sketching

Curve Sketching

Name _____

SA1

Sketch each quadratic curve from its roots and turning point.



Sketch y = (x + 2)(x - 2). State the roots and the turning point.

Show your working

KAIZEN MATHS

Lesson starter

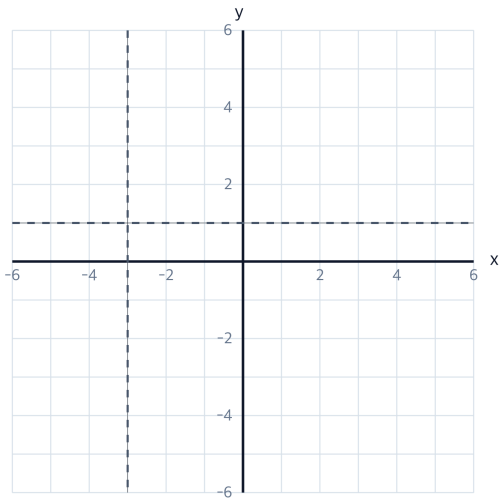
Graph Transformations and Curve Sketching

Curve Sketching

Name _____

SB1

Sketch each reciprocal graph and state the asymptotes.



Sketch y = $\frac{2}{x+3}$ + 1. State the vertical and horizontal asymptotes.

Show your working

KAIZEN MATHS

Lesson starter

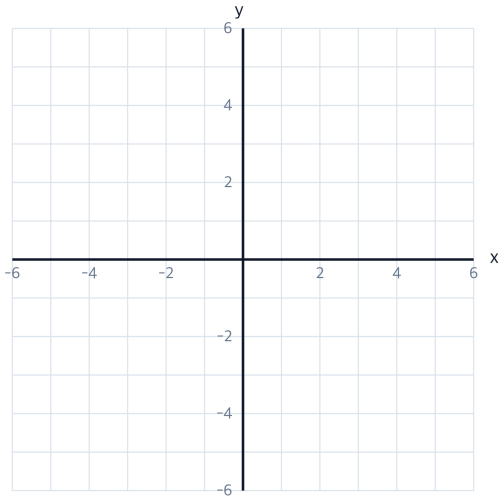
Graph Transformations and Curve Sketching

Curve Sketching

Name _____

SC1

Sketch each cubic curve using its roots and end behaviour.



Sketch the cubic curve y = (x + 1)(x - 1)(x - 4). State the roots and end behaviour.

Show your working

KAIZEN MATHS

Lesson starter

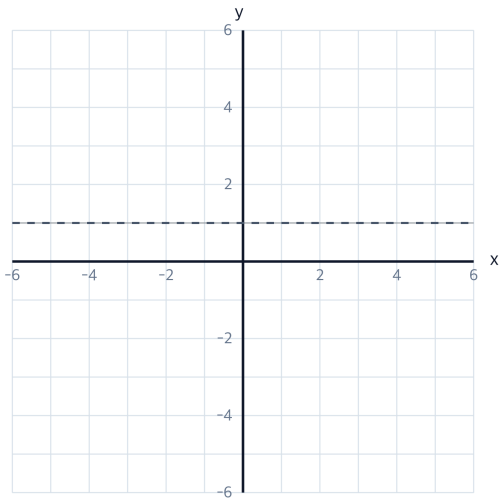
Graph Transformations and Curve Sketching

Curve Sketching

Name _____

SD1

Sketch each exponential curve and state the horizontal asymptote.



Sketch $y = 2^{x+2} + 1$. State the horizontal asymptote and three useful points.

Show your working

KAIZEN MATHS

Exit ticket

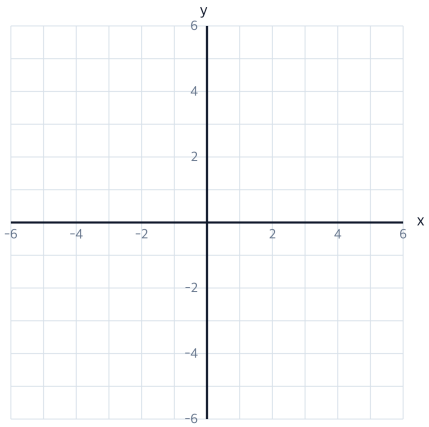
Graph Transformations and Curve Sketching

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Name _____

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KAIZEN MATHS

Exit ticket

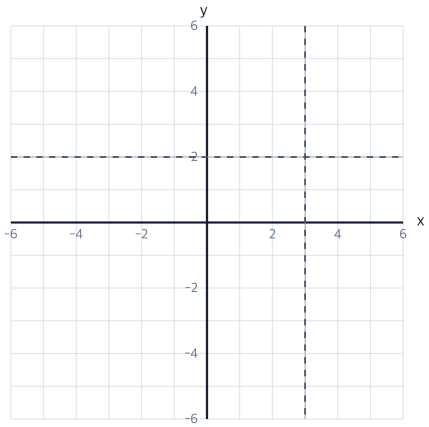
Graph Transformations and Curve Sketching

Curve Sketching

Name _____

EB1

Sketch each reciprocal graph and state the asymptotes.



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KAIZEN MATHS

Exit ticket

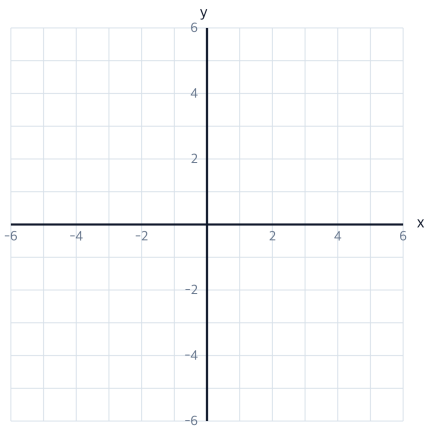
Graph Transformations and Curve Sketching

Curve Sketching

Name _____

EC1

Sketch each cubic curve using its roots and end behaviour.



Sketch the cubic curve y = (x + 2)(x - 0)(x - 3). State the roots and end behaviour.

Show your working

KAIZEN MATHS

Exit ticket

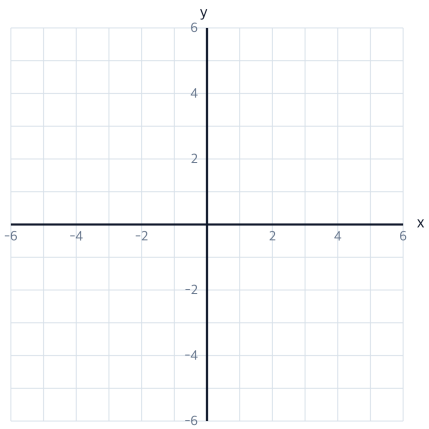
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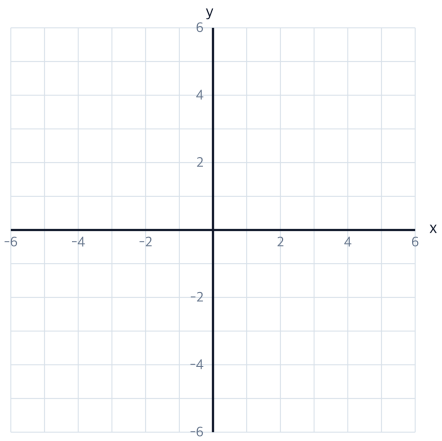
Teacher answers & working

Graph Transformations and Curve Sketching · Curve Sketching

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SA1

Sketch each quadratic curve from its roots and turning point.



Sketch y = (x + 2)(x - 2). State the roots and the turning point.

Answer:

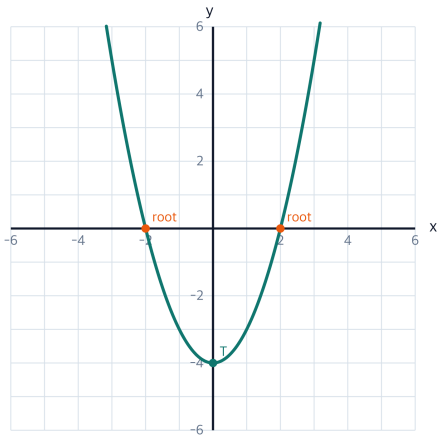
Roots: x = -2, x = 2. Turning point: (0, -4).

Working:

In factorised form, each bracket gives a root: x = -2 and x = 2.

The axis of symmetry is halfway between the roots: x = (-2 + 2) ÷ 2 = 0.

Substitute x = 0 into the equation to find the turning point: y = -4.



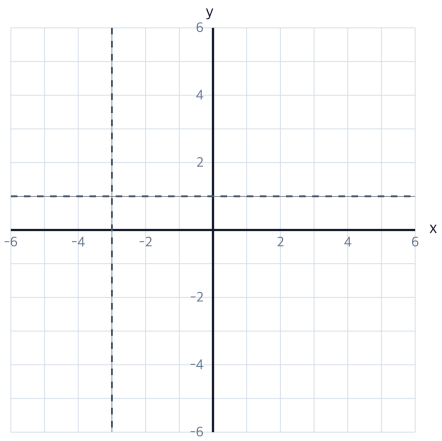
Teacher answers & working

Graph Transformations and Curve Sketching · Curve Sketching

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SB1

Sketch each reciprocal graph and state the asymptotes.



Sketch $y = \frac{2}{x+3} + 1$. State the vertical and horizontal asymptotes.

Answer:

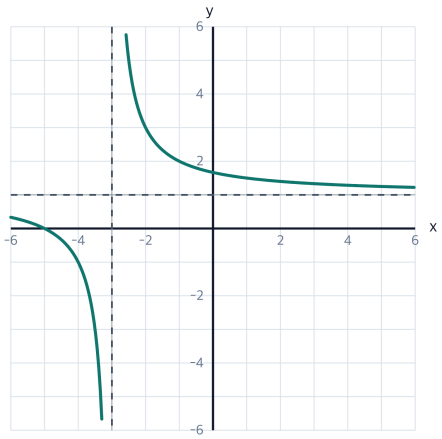
Vertical asymptote: x = -3. Horizontal asymptote: y = 1.

Working:

The denominator is zero when x + 3 = 0, so the vertical asymptote is x = -3.

The graph is translated vertically by 1, so the horizontal asymptote is y = 1.

The sign and size of the numerator, 2, decide the branch direction and steepness.



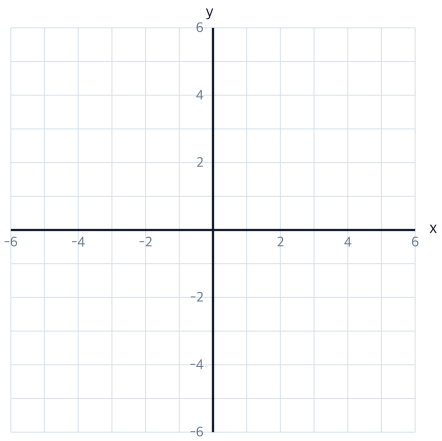
Teacher answers & working

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SC1

Sketch each cubic curve using its roots and end behaviour.



Sketch the cubic curve y = (x + 1)(x - 1)(x - 4). State the roots and end behaviour.

Answer:

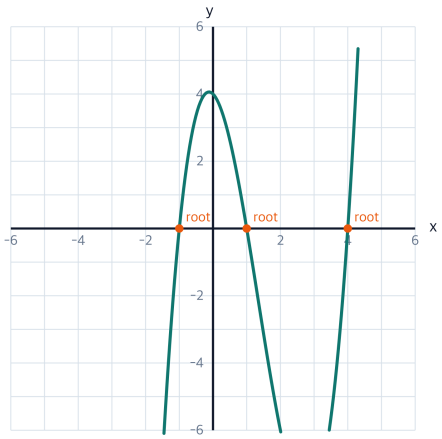
Roots: x = -1, x = 1, x = 4. Falls left and rises right.

Working:

Each factor gives a root: x = -1, x = 1, x = 4.

The coefficient is positive, so the curve falls on the left and rises on the right.

Sketch the curve crossing the x-axis at each root.



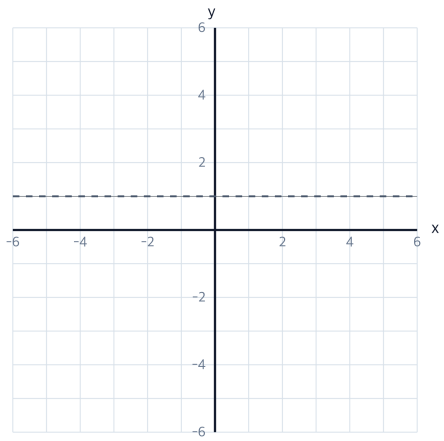
Teacher answers & working

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SD1

Sketch each exponential curve and state the horizontal asymptote.



Sketch $y = 2^{x+2} + 1$. State the horizontal asymptote and three useful points.

Answer:

Horizontal asymptote: y = 1. Useful points include (-2, 2), (-1, 3), (0, 5).

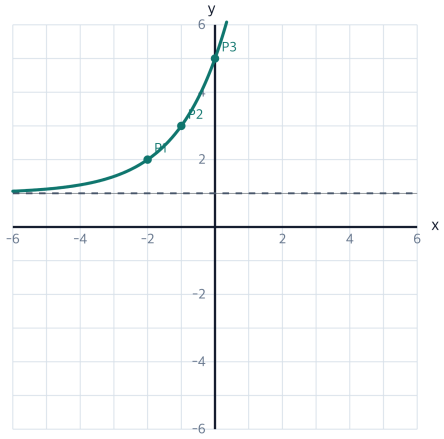
Working:

The parent graph $y = 2^x$ has horizontal asymptote y = 0.

The + 1 outside the power moves the asymptote to y = 1.

The expression in the power shifts the graph left by 2.

Map parent points to get useful image points: (-2, 2), (-1, 3), (0, 5).



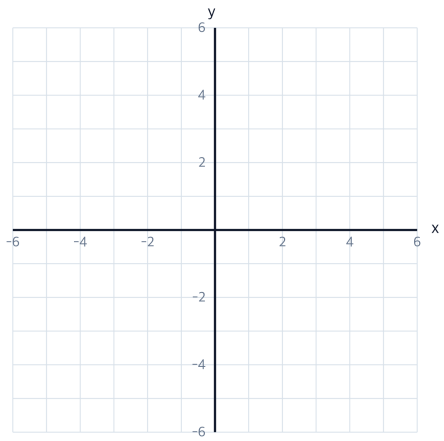
Teacher answers & working

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EA1

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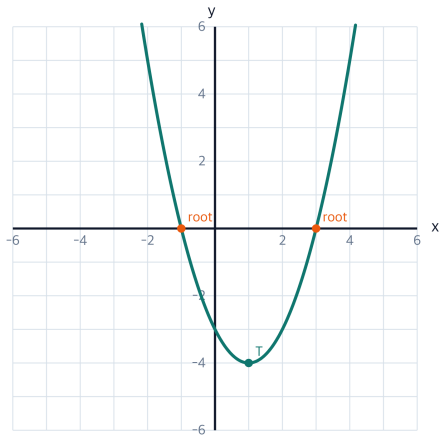
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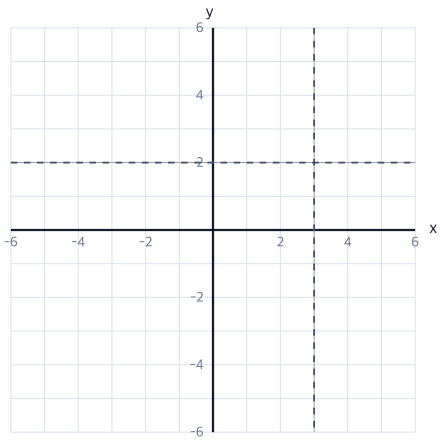
Teacher answers & working

Graph Transformations and Curve Sketching · Curve Sketching

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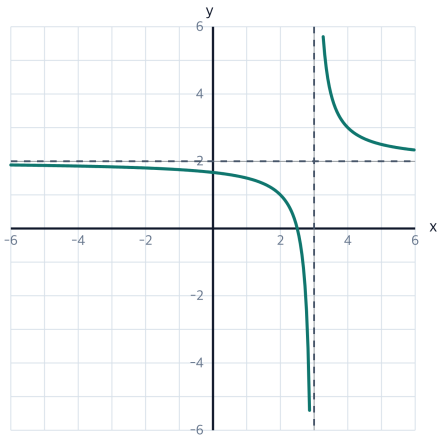
Vertical asymptote: x = 3. Horizontal asymptote: y = 2.

Working:

The denominator is zero when x - 3 = 0, so the vertical asymptote is x = 3.

The graph is translated vertically by 2, so the horizontal asymptote is y = 2.

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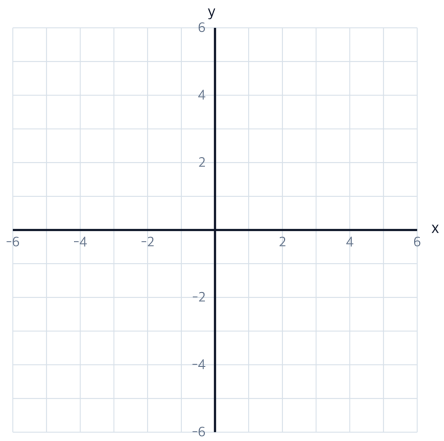
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